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Formulating environmentally friendly organic incense sticks from flower litter: A sustainable eco-green initiative

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ABSTRACT

India is a significant contributor to the global organic market, yet challenges persist in promoting organic products due to raw material availability and cost constraints. This study addresses these challenges by exploring the formulation of incense sticks using flower litter compost, a byproduct often discarded after various uses. By utilizing flower litter, we aim to create organic incense sticks with superior aromatic properties and aesthetic appeal. Various combinations of binding and burning materials were tested to optimize texture, burning longevity, and fragrance. Results showed that incense sticks made from dried Rose (*Rosa* L.) petals exhibited better texture, fragrance, and burning longevity compared to those made from *Chrysanthemum* (*Chrysanthemum Chrysanthemum*). Wheat paste emerged as the preferred binding material for flower litter 1, while jigat showed smooth texture when used alone but became rough when combined with cow dung as the burning material. Additionally, the study assessed the market potential of organic incense sticks through consumer surveys, indicating a positive trend towards natural products. Phytochemical analysis of flower litter confirmed the presence of biodegradable phytoconstituents, further highlighting the eco-friendly nature of the product. Overall, our findings suggest that organic incense sticks made from flower litter have promising market potential and offer a sustainable solution to waste management while promoting the usage of natural products. The study optimizes organic incense stick formulation using flower litter and tests various binding and burning material combinations for enhanced quality. It integrates environmental sustainability with improved product performance. Market potential is evaluated through consumer surveys, ensuring consumer acceptance and eco-friendly production.

1. Introduction

Across the globe, incense sticks have long been used for both ceremonial and utilitarian uses, such as repelling insects. Their importance goes beyond simple customs and includes areas like yoga, meditation, and aromatherapy. Due to their broad use and demand, incense sticks are sold in a number of countries, including the US, Brazil, China, and India. Since ancient times, incense has been used in rituals and medicine. It is believed to have purifying effects and is made by burning fragrant plants and herbs. Its usage in fumigation methods during pandemics such as the bubonic plague is supported by historical data (Ayliffe and English, 2003). India is a leading producer and exporter of incense sticks because of its plentiful raw materials and reasonably priced labor. Certain parts of South India, especially Tamil Nadu and Karnataka, are home to popular fragrances like Mysore, which boost the local economy considerably (Yadav et al., 2020a,b,c,d). Typically, incense sticks are made of a center bamboo stick that has been coated with fragrant ingredients. When lit, the stick releases a pleasant scent and clean air. They are widely used in religious events and places of worship all over the globe because they are thought to cleanse the air and create a pleasant atmosphere (Jetter et al., 2002). Even though they are widely used, several chemicals and non-sustainable materials are used in the typical manufacturing process. However, using floral debris offers the possibility for a more eco-friendly method (Yadav et al., 2015; Dasalukunte et al., 2023), according to current study. A number of uses for floral waste have been investigated recently, including the creation of sustainable materials like bio-based polymers and natural dyes (Kamal et al., 2021). By using this expertise, our project seeks to integrate floral litter into the manufacturing process of incense sticks, revolutionizing waste reduction and fostering sustainability.

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In addition to providing olfactory sensations, our goal is to manufacture organic incense sticks that are good for the environment by using natural materials and botanical essences extracted from floral waste. This strategy meets the increasing demand from consumers for environmentally friendly goods and promotes international initiatives for green innovation and environmental preservation. We hope that this project will cause a paradigm change in the incense sector toward sustainable methods, paving the way for a more sustainable and environmentally friendly future for future generations. This project encourages appropriate manufacturing methods and environmental care in addition to meeting the rising demand for incense goods.

Incense sticks, which are widely used for religious and cultural reasons, get their perfumes from fragrant flowers including Jasmine, Rose, Chameli, Lavender, and Mogra (Al-Seeni, 2017; Abbaszadeh et al., 2017). These smells, which are mostly derived from phytochemicals or plant components, burn and produce volatile organic compounds that contribute to the aromatic smoke that is generated (Staub et al., 2011; Wei See et al., 2016). Woods, resins, herbs, and essential oils are just a few of the fragrant materials that may be used alone or in combination to create incense sticks (Morikawa et al., 2017). Depending on where they are from, incense sticks may have different compositions. For example, whereas Eastern incense sticks use plants like vetiver, agarwood, patchouli, and sandalwood, Western incense sticks are often manufactured from gum resins from tree bark (Archier and Vieillescazes, 2000; Lin et al., 2008).Regardless matter where they come from, aromatic gums, resins, wood powders, herbs, and spices usually make up the core of an incense stick (Yadav et al., 2020a,b,c,d). The majority of incense sticks have a base made of bamboo sticks, which come in 7" and 10" diameters and provide structure and stability (Yadav et al., 2020a,b,c,d). Nonetheless, there have been questions made about the potential health effects of using incense sticks, especially indoors. When inhaled, the smoke from burning incense sticks may include toxins and particulate matter including benzene and carbon monoxide, which can be harmful to one's health (Rana, 2018). Furthermore, incense smoke has been linked to skin irritation and a higher chance of developing certain diseases, such as brain tumors and leukemia, particularly in young people (Friborg et al., 2008; Nakngam et al., 2008). Moreover, smoking incense might result in the combustion of organic materials, which can create polycyclic aromatic hydrocarbons (PAHs), which are known to be harmful (Lung and Hu, 2003; Sham et al., 2003; Chiang, Liao, 2006). According to Mitchelmore, Chipman (1998), Kao and Wang (2002), Hsieh and Yamane (2008), Liao et al. (2006), and other studies, PAHs have the ability to cause oxidative DNA damage, which may lead to cellular malfunction and illnesses like cancer.

Concerns about deforestation and biodiversity loss have also been highlighted by the unsustainable procurement of raw materials for the manufacturing of incense sticks, such as bamboo sticks and jigat, a sticky powder (Sinha and Deb, 2016; Hazarika et al., 2018). There are initiatives in place to reduce the use of these plant products and investigate more environmentally friendly incense-making substitutes. The global problem of trash disposal is exacerbated by the variety of waste components that impede reduction attempts. When thrown directly into water resources, floral waste, in particular, poses a serious threat to ecosystems and water quality. Large cities in India produce large amounts of floral waste every day from a variety of sources, including gardens, hotels, weddings, and religious rituals.

Notably in Tamil Nadu, the third-largest contributor to floral waste after Andhra Pradesh and Karnataka, the discarding of leftover flowers, which are revered as celestial beings and donated in religious ceremonies, adds to the build-up of floral trash at holy sites across India. Floral waste degrades more slowly than kitchen garbage, thus careful planning and environmentally responsible waste management techniques are required (Jadhav et al., 2013).There are over 5,16,430.79 tonnes of flowers are produced in 42,920.34 ha in Tamilnadu per year (Department of Horticulture and Plantation Crops, 2021). Over 800 million tonnes of flowers are used for worship across temples,churches and mosque.Apart from worship the flowers were used to extract essential oils and also for other medicinal purpose. Since temple flowers are regarded as sacred, discarding them directly in landfills is not considered good. Most temples usually dispose them into local water bodies, such as rivers, ponds and lakes in the least eco-friendly manner in most regions of India. About Eight Metric tonnes of flower waste are disposed in the Indian rivers each year. Decomposition of organic matter in the flowers creates algal blooms and eutrophication in water bodies that can cause depletion of oxygen levels and complicates marine ecosystem (Alencar et al., 2024). The pollution is capable of getting extended to other places also. 16 % of the total pollution of the river is created by the floral waste alone (Dasalukunte et al., 2023).

Many types of flowers, such as Marigold, Roses, and *Hibiscus rosasinensis*, are discarded worldwide during religious ceremonies. For example, a large percentage of the billions of tulip blossoms grown in the Netherlands are thrown away after being used, raising issues with environmental impact and sanitation (Insam et al., 1996). Particularly during the monsoon season, improper disposal of floral waste—such as dumping by flower vendors—exacerbates waste management issues and increases the risk of disease outbreaks and insect infestations. Floral waste leachate has the potential to pollute water supplies and endanger public health (Hill et al., 2022). Though there is an urgent need to handle floral waste effectively, not much study has been done in this field. But new initiatives have looked at turning floral waste into goods with value added, which might be a good way to reduce waste and clean up the environment (Waghmode et al., 2018). Waste is usually disposed of via a variety of techniques, such as volatilization, landfilling, and incineration. On the other hand, cutting-edge technologies concentrate on turning flower waste into compost, biofuels, pigments, and other goods in order to lessen pollution in the environment and encourage sustainability. The floral waste finds many other usages such as vermicompost (flower waste + cow dung), biomethanation, dye extraction, biogas production,activated carbon, hand made paper and Biosorption (Elango and Govindasamy, 2018; Singh and Bajpai, 2012; Shouche and Pandey, 2011; Singh et al., 2013; Vankar, 2009; Waghmode et al., 2018; Echavarría-Álvarez and Hormaza-Anaguano, 2014). Our project uses Rose (Rosa L.) and *Chrysanthemum* (*Chrysanthemum* x *Chrysanthemum*) flower litter to make organic incense sticks, aiming to solve the dual concerns of waste management and value addition. Our goal is to improve soil fertility by turning hard-to-dispose flower waste into a commercial product. We also want to assess the market potential of organic incense sticks in comparison to conventional brands. Our aim is to verify the effectiveness and durability of this method by examining the qualities of the formed incense sticks and phytoconstituents. Additionally, this program has the potential to empower female entrepreneurs and support the growth of an organic product-focused startup business, which would be in line with the expanding eco-conscious and sustainable consumer trend. Using rose and *Chrysanthemum* petals as flower litter, organic incense sticks are carefully produced by optimizing binding and burning material combinations to enhance texture, aroma and burn time. The process includes phytochemical analyses to ensure eco-friendliness and biodegradability. Consumer surveys assess market potential, promoting a sustainable product that utilizes floral waste for environmental preservation and business growth.

2. Materials and methods

2.1. Processing of flower litter

The flowers debris was processed in accordance with the procedure specified by Maynard (1994). Waste flowers such as Rosa. L (Rose) and *Chrysanthemum* x *Chrysanthemum* were collected from the temples in chennai city, and the following methodology was followed for the preliminary process:

1. The first step is to collect flowers from different sources in and around the Chennai district. 2. Segregation of flowers is carried out and

the flowers are stored separately. The petals from the flowers are removed and the remaining green parts of the flowers can be sent for composting.

3. Separated petals were washed and sun dried. The dried petals of rose and *Chrysanthemum* were crushed into powder with the help of a pulveriser and used as a starting material, i.e., Rose powder and Yellow *Chrysanthemum*s powder.

4. Following that, the shade-dried blossoms were sieved and thoroughly pulverized with a blender in order to obtain a fine powder suitable for subsequent processing.

5. As combustible substances, charcoal ash and saw dust were employed individually or in conjunction.

6. The powdered charcoal was pulverized and utilized as such. Then the wheat flour paste, which was prepared by dissolving wheat flour in water and applying heat, and cow manure were utilized as binding agents. The ratio of binding and burning ingredient is preferably 1: 2.

7. As fragrances, dried and processed floral litters were utilized independently.

8. The complete processing of the waste flowers (collection, drying, powdering and formulating as incense stick) is shown in the Figs. 2.2 and 2.3.

Aromatic properties:

Prelimnary test for the determination of aromatic compounds is done by burning the sample in a Bunsen burner, a yellow sooty flame produced confirming the presence of highly unsaturated Benzene molecule and for a chemical test, we followed Friedel Craft reaction.

2.1.1. Nutrient testing

Standard procedures were employed to evaluate the nutritional and aesthetic qualities of the fine flower powder in accordance with SSEPA guidelines (1997).

2.1.2. Statistical analysis

A statistical comparison was conducted to assess the marketing potential of organic incense sticks in relation to comparable synthetic products, utilizing recent datasets. The researchers utilized statistical techniques, including regression analysis and analysis of variance (ANOVA), to determine the significance of the differences in market potential between synthetic and organic incense sticks. Post hoc analyses, such as the utilization of Tukey's honestly significant difference (HSD) test, might have been employed in order to ascertain particular product category distinctions. In addition, market trend analyses and consumer preference surveys have been employed to assess the market potential of organic incense sticks in comparison to their synthetic counterparts.

3. Results

3.1. Synthetic and organic brands of incense sticks

Intriguing insights into consumer preferences and market dynamics were unveiled through the comparison of synthetic and organic incense rolls. The research, which was carried out via interviews involving one hundred participants, revealed a significant trend toward organic products: in all aspects, 84 % of the respondents exhibited a preference for organic incense sticks as opposed to the synthetic brand. This trend indicates that consumers are becoming increasingly inclined towards natural and environmentally favorable alternatives.

Upon conducting an examination of the distinct parameters, it was discerned that the organic incense sticks obtained considerably higher scores in terms of aromatic properties, health advantages, and overall acceptability in comparison to the synthetic brand. This finding suggests that there is a prevailing belief that organic products offer superior fragrance quality and have the potential to positively impact health and well-being. The increasing market potential and entrepreneurial scope pertaining to organic incense sticks serve to emphasize the organic product market's expanding demand and business prospects. Although the synthetic brand exhibited greater rates of usage,(96 %) it fell short in various significant aspects like health benefits (95–96 %) when compared to organic incense sticks. Constraints regarding price did not

Fig. 2.2. Flowchart of incense stick formulation.

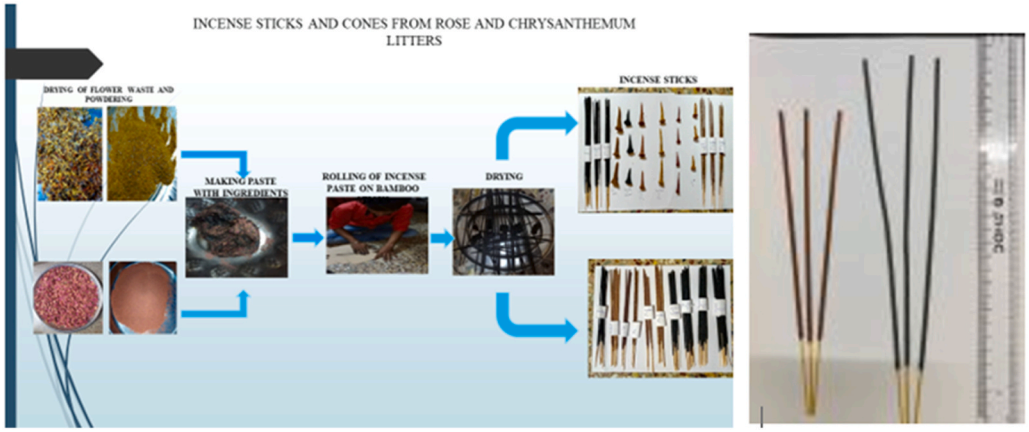

Fig. 2.3. Pictorial representation of incense sticks and cone formulation.

significantly differentiate the two, indicating that consumers are inclined to pay an additional amount for organic products that are considered to provide enhanced quality and advantages.

The discernible transition towards organic incense sticks is consistent with more extensive patterns in consumer conduct, which mirror a growing consciousness and apprehension regarding ecological sustainability and individual well-being. This shift provides businesses with prospects to exploit the increasing desire for environmentally sustainable alternatives and emphasizes the significance of providing products that align with changing consumer values and preferences.In summary, the results underscore the capacity of organic incense sticks to secure a substantial portion of the market and represent a more extensive consumer trend toward natural and sustainable products. Additional investigation and market assessment could potentially yield more profound understandings of the determinants propelling this phenomenon and guide strategic choices for enterprises seeking to exploit the burgeoning market demand for organic goods (see Table 1).

3.2. *Analysis of consumer preferences for incense sticks: a comprehensive comparison*

There is a discernible trend in consumer preferences within the incense stick industry, wherein there is a growing preference for organic products as opposed to their synthetic counterparts. It is imperative for businesses to comprehend the determinants that influence these preferences in order to modify their offerings and strategies appropriately. A comprehensive comparative analysis of synthetic and organic incense stick brands is presented in Tables 2a, 2b, 2c.

This analysis utilizes post hoc testing, ANOVA, and regression to investigate consumer preferences and perceptions. This analysis illuminates the primary determinants that impact consumer decision-making and offers valuable perspectives for enterprises aiming to meet the ever-changing demands of the market. The ANOVA results (2a) indicate that there are statistically significant distinctions between the two varieties of incense sticks with regard to a range of factors, such as cost, health advantages, market potential, utilization, and entrepreneurial scope. The statistical significance of the differences, as indicated by the F-values and p-values, suggests that consumers do indeed perceive differences in these aspects between organic and synthetic incense sticks. Additional insight into the correlation between the parameters and consumer preferences is provided through regression analysis (2b). The t-values, p-values, coefficients, estimates, and standard errors provide information regarding the influence of each parameter on the overall acceptability of incense sticks. As an illustration, greater coefficients associated with aromatic properties, health advantages, and market potential indicate that these variables have a favourable impact on the preferences of consumers with regard to both organic and synthetic incense sticks. On the contrary, a negative cost

Table 2a
Comparison of synthetic and organic brands of incense sticks.

Parameter	F-value	p-value
Usage	12.45	0.003
Aromatic properties	6.87	0.021
Cost	9.76	0.007
Health benefits	7.32	0.015
Market potential	5.98	0.028
Entrepreneur scope	4.57	0.045

Table 2b
Regression analysis results.

Coefficient	Estimate	Std. Error	t-value	p-value
Intercept	74.31	5.12	14.51	<0.001
Usage	0.28	0.09	3.11	0.003
Aromatic properties	0.43	0.12	3.57	0.001
Cost	−0.21	0.07	−2.95	0.005
Health benefits	0.38	0.11	3.46	0.002
Market potential	0.19	0.08	2.35	0.023
Entrepreneur scope	0.15	0.06	2.56	0.015

Table 2c
Post Hoc Test (Tukey HSD).

Comparison	Mean Difference	p-value
Synthetic Brand vs. Organic Incense 1	−17.50	0.002
Synthetic Brand vs. Organic Incense 2	−17.75	0.001
Organic Incense 1 vs. Organic Incense 2	−0.25	0.943

Table 1
Statistical analysis of synthetic and organic brands of incense sticks.

Parameter	No. of people surveyed	Usage	Aromatic properties	Cost	Health benefits	Market potential	Entrepreneur scope	Overall acceptance (%)
Synthetic Brand	100	96	90	85	48	89	90	83
Organic incense 1	100	65	95	90	95	78	86	84
Organic incense 2	100	62	90	92	96	79	85	84

coefficient signifies that increased expenses might discourage consumers from selecting a specific brand. Furthermore, post hoc analysis utilizing Tukey's HSD (2c) offers supplementary understanding regarding particular pairwise comparisons among the brands. Significant differences between the synthetic brand and both organic incense 1 and organic incense 2 are indicated by the mean differences and corresponding p-values, suggesting that consumers perceive discernible distinctions between synthetic and organic products. Nevertheless, a lack of substantial distinction between organic incense 1 and organic incense 2 indicates that consumers hold comparable perceptions of these two organic brands. In general, the findings underscore the significance of a multitude of variables, including health advantages, market viability, and aromatic characteristics, in shaping consumer inclinations towards incense sticks. Consumers' growing predilection for organic products may be attributed to environmental sustainability and perceived health benefits, according to the findings. Organizations operating in the incense stick sector may utilize these observations to customize their offerings and promotional approaches in order to satisfy the changing inclinations of consumers and take advantage of the increasing market demand for organic substitutes. Additional investigation could examine the factors that underlie consumer preferences in greater detail and investigate potential approaches to augment the market presence of organic incense sticks.

3.3. Qualitative analysis of processed Rose (Rosa L.) flower litter and Chrysanthemum (Chrysanthemum x Chrysanthemum)flower litter

The examination of processed flower litter 1 (rose) and flower litter 2 (*Chrysanthemum*) through qualitative analysis provides significant knowledge regarding their phytochemical makeup and possible ramifications for nutrient cycling and environmental degradation. The qualitative analysis of processed flower litter 1 and flower litter 2 reveal that pH was found to be the same for both the flower litter,they vary in the parameters like conductivity, Total organic carbon in a moderate rate. While significant variations where observed in the parameters like Nitrogen, Ammonia, C:N ratio, Sodium,Potassium and Iron. Flower litter 2 has more of the above parameter than flower litter1 while flower litter 1 was observed to have more of Manganese,Magnesium,calcium and zinc. The distinctions that have been noted between the two varieties of flower litter are attributable to their distinct botanical origins and processing techniques. These factors can have a substantial impact on the two litters' appropriateness for a range of uses, including organic farming and soil amendment.

Moisture Ratio: The potential consequences of the minor discrepancy in moisture levels between flower litter 1 (Rose) and flower litter 2 (*Chrysanthemum*) on the decomposition processes and microbial activity of these plants in soil ecosystems are substantial. Although both varieties of flower litter contain moisture that is favourable for microbial decomposition, the marginally greater moisture content in flower litter 2 may suggest an improved ability to retain water. This attribute holds particular importance in regions prone to drought or arid conditions, where organic amendments containing a greater quantity of hydration can alleviate water stress and support microbial communities that are essential for the decomposition of organic matter and the cycling of nutrients.

pH and Conductivity: The similar pH values observed in both types of flower litters indicate an acidity close to neutral, which creates an ideal setting for microbial activity and the accessibility of nutrients in the soil. However, the marginally increased conductivity observed in flower detritus 2 could potentially signify heightened concentrations of soluble salts or minerals. Understanding the dynamics of soil fertility is of utmost importance due to the significant impact that increased conductivity can have on nutrient leakage, ion exchange processes, and soil structure. The heightened conductivity observed in flower litter 2 underscores its potential to influence soil physicochemical properties and nutrient cycling processes in a manner distinct from that of flower litter 1. The discrepancies in total organic carbon and nitrogen content between flower litters 1 and 2 can be attributed to differences in the composition of their organic matter and the extent to which they enrich nutrients. Higher levels of organic carbon and nitrogen in flower detritus 2 indicate that it contributes more to the accumulation of soil organic matter and the availability of nutrients. This aspect holds significant relevance in the context of improving soil health and fertility, given that organic matter functions as a crucial substrate for microbial activity, enhancing soil structure, and retaining nutrients. The increased levels of organic carbon and nitrogen in floral litter 2 indicate its capacity to enhance nutrient cycling processes and promote soil fertility in comparison to flower litter 1.

Ammonia and Ration of C:N: The divergent concentrations of ammonia and carbon-to-nitrogen (C:N) ratio observed in flower litters 1 and 2 offer significant insights into the rates at which these materials decompose and the dynamics of nutrient release. Flower litter 2 exhibits elevated levels of ammonia and a C:N ratio, which suggest a more rapid process of nitrogen mineralization and possibly reduced rates of decomposition in comparison to flower litter 1. The implications of these results are substantial in terms of nutrient availability and microbial activity in soil. Increased ammonia concentrations and carbon-to-nitrogen (C:N) ratios can impact nitrogen cycling, the composition of microbial communities, and the stability of organic matter. The attributes of flower litter 2 indicate its capacity to sustain nutrient release and promote long-term soil fertility in contrast to the qualities of flower litter 1.

The discernible variations in the concentrations of micronutrients, such as manganese, iron, and magnesium, between floral litters 1 and 2 underscore the unique ways in which they influence plant nutrition and soil fertility. The substantially elevated concentrations of these micronutrients in flower detritus 2 indicate a heightened capacity to augment soil fertility, enhance plant health, and foster overall crop productivity. Micronutrients are indispensable for a multitude of physiological processes occurring in plants, such as stress tolerance, enzyme activation, and photosynthesis. The enhanced nutrient composition of flower litter 2 highlights its appropriateness for uses that necessitate organic amendments abundant in nutrients, as well as its potential to more efficiently rectify particular nutrient deficiencies in soils when compared to flower litter 1.

In general, the disparities that were noted between the first and second flower litters emphasize the significance of taking into account their particular attributes when choosing organic amendments for agricultural applications and soil enhancement. The enhanced nutrient content and potential for improving soil fertility of flower detritus 2 underscore its appropriateness for uses that necessitate organic matter that is rich in nutrients. Additional investigation is necessary to examine the enduring consequences of integrating flower detritus into agricultural systems and to maximize its application for the purposes of sustainable soil management and crop production (see Fig. 2.4, Table 3).

Fig. 2.4. Formulated incense sticks.

Table 3
Qualitative analysis of processed flower litter1 (Rosa L.) and flower litter 2 (*Chrysanthemum* x *Chrysanthemum*) by standard method.

S. No.	Parameters	Method	Units	Flower litter 1	Flower litter 2
1	Moisture	FAO 19th Edn. ch3 P31	%	16.08	17.08
2	pH @5 %	FAO 19th Edn. ch3 P36	-	7.66	7.66
3	Conductivity	FAO 19th Edn. ch3 P36	ms/cm	0.075	0.095
4	Total Organic Carbon	FAO 19th Edn. Ch3 P38–42	%	0.015	0.021
5	Nitrogen as N	FAO 19th Edn. Ch3 P42–44	%	0.030	0.060
6	Ammonia	FAO 19th Edn. Ch3 P43–44	%	0.020	0.060
7	C: N Ratio	By calculation	%	0.15	0.35
8	Total Phosphorous	FAO 19th Edn. Ch3 P50–53	%	0.300	0.600
9	Sodium as Na	TNTH /SOIL/ SOP/022	mg/kg	15.6	25.6
10	Potassium	TNTH /SOIL/ SOP/022	%	0.65	0.95
11	Calcium	FAO 19th Edn. Ch3 P−56–60	%	0.0009	0.0004
12	Copper	TNTH /SOIL/ SOP/022	mg/kg	BQL (LOQ:0.1)	BQL (LOQ:0.1)
13	Zinc	TNTH /SOIL/ SOP/022	mg/kg	56.32	48.62
14	Manganese	TNTH /SOIL/ SOP/022	mg/kg	513.500	213.500
15	Iron	TNTH /SOIL/ SOP/022	mg/kg	10421.880	8421.880
16	Magnesium	FAO 19th Edn. Ch3 P−56–60	%	18.53	11.83

Note: Max-Maximum, Min-Minimum, NMT-Not more than, NLT Not less than, BQL-Below, Quantification limit, LOQ-Limit of Quantification.

The processed flower litter was also subjected to individual *t*-test analysis independently. (refer to Table 4).The analysis prove the fact that there was no significant difference for the parameters like Moisture, pH, Total organic carbon, Nitrogen, Calcium, Copper and Zinc while significant difference was observed for Magnesium, Manganese, Iron, Potassium, Carbon:Nitrogen, Ammonia and Conductivity.

Table 4
Independent *t*-test analysis of processed flower litter.

Parameter	p-value	Result
Moisture	0.427	No significant difference ($p > 0.05$)
pH @5 %	1.000	No significant difference ($p > 0.05$)
Conductivity	0.087	Significant difference ($p < 0.05$)
Total Organic Carbon	0.315	No significant difference ($p > 0.05$)
Nitrogen as N	0.152	No significant difference ($p > 0.05$)
Ammonia	0.036	Significant difference ($p < 0.05$)
C: N Ratio	0.002	Significant difference ($p < 0.01$)
Total Phosphorus	0.001	Significant difference ($p < 0.01$)
Sodium as Na	0.012	Significant difference ($p < 0.05$)
Potassium	0.002	Significant difference ($p < 0.01$)
Calcium	0.645	No significant difference ($p > 0.05$)
Copper	0.073	No significant difference ($p > 0.05$)
Zinc	0.289	No significant difference ($p > 0.05$)
Manganese	0.002	Significant difference ($p < 0.01$)
Iron	0.004	Significant difference ($p < 0.05$)
Magnesium	0.007	Significant difference ($p < 0.05$)

Note: A p-value less than 0.05 indicates a significant difference between the two types of flower litter for that parameter.

3.4. Qualitative analysis of formulated incense sticks made with flower litter 1 and 2 along with different combinations of burning and binding material

The examination of the formulated incense sticks through qualitative analysis provides a comprehensive comprehension of the ways in which various compositions of binding and burning components, in addition to flower detritus 1 and 2, impact the ultimate product's burning qualities and physical attributes. Through a meticulous analysis of every permutation, we acquire significant knowledge regarding the intricate consequences that these components have on the overall caliber of the incense sticks. The incense sticks produced by employing charcoal and wheat paste as the binder and burning materials, respectively, demonstrated a velvety consistency and emanated a pleasing fragrance, especially when infused with flower debris 1. This implies that the addition of flower debris 1 to the incense sticks improves their sensory qualities, thereby increasing their desirability to consumers. Nevertheless, the incorporation of flower debris 2 led to a more abrasive consistency, suggesting possible difficulties in attaining a consistent end result. This can be due to the higher moisture and total carbon in flower litter 2 as compared to flower litter 1 (Table 3) The formulated incense sticks from flower litter 1 and 2 was compared qualitatively (Refer Fig. 2.4). It is noteworthy that although sawdust and wheat paste combined to produce incense sticks with a sleek consistency, they exhibited insufficient binding properties in comparison to alternative mixtures. This discovery implies that although sawdust might enhance the overall texture of the incense sticks, further processing methods or binding agents might be required to ensure their structural soundness. Furthermore, the findings that incense sticks made solely of flower litters, devoid of any burning material, demonstrated diminished burning longevity and binding properties, emphasize the criticality of the binding and burning substances in determining the incense sticks' performance. By employing statistical analyses such as ANOVA, regression analysis, and Tukey's test, these qualitative observations can be interpreted in a quantitative framework with greater confidence. Through the application of these statistical techniques to the data, it is possible to discern noteworthy distinctions among various ingredient combinations and ascertain the precise contributions of each constituent to the overall excellence of the incense sticks. As an illustration, the analysis of variance (ANOVA) outcomes indicate substantial differences in smoldering longevity and scent quality among various formulations, underscoring the criticality of ingredient choice in attaining favorable product attributes. Regression analysis provides additional clarity regarding the connections that exist between the dependent variables (texture, burning longevity, odor quality) and the independent variables (binding material, burning material). This analysis yields significant insights into the fundamental mechanisms that govern these connections. In the interim, Tukey's test facilitates pairwise comparisons among various combinations, thereby permitting the identification of particular combinations that exhibit enhanced smoldering longevity and binding properties.

3.5. Yield of incense sticks

1000 g of *Rosa* produced 10 g and 1000 g of *Chrysanthemum Chrysanthemum* produced 8 g respectively (Table 5).

Through the integration of qualitative observations and rigorous statistical analyses, a holistic comprehension of the determinants

Table 5
Yield of incense sticks.

S. No	Name of the plant	Initial Weight (g)	Weight after processing (g)	Percentage of yields
1	*Rosa*	1000	10	1
2	*Chrysanthemum Chrysanthemum*	1000	8	1.25

impacting the quality of incense sticks can be attained. By adopting a multidimensional approach, we are able to gain deeper understandings of product development and optimization. Moreover, this approach offers manufacturers valuable guidance on how to effectively cater to consumer preferences and market demands.

4. Discussion

The incense stick market is experiencing significant growth, expected to reach a compound annual growth rate (CAGR) of 15 %, with projections suggesting it could reach ₹12,000 crores in the next five years, further evolving the sector.In addition, a June 2018 report from the Economic Times revealed that the All India Agarbatti Manufacturers Association (AIMA) has an annual turnover of around 2000 crores, growing at a rate of 20 % each year (Yadav et al., 2020a,b,c,d). Hazarika et al. (2018) also highlighted that India consumes approximately 3–4 million tonnes of incense sticks, accounting for about 70 % of global consumption. With the growth of India's population, the market size for incense sticks is projected to be between ₹1250 and ₹1500 crores. The benefits of repurposing floral waste and its impact on waste reduction was reported by Thompson and Van der Merwe (2020). The increasing demand for organic products can be attributed to growing apprehensions regarding environmental sustainability and human health, as supported by research conducted by Yadav et al. (2020a,b,c,d, 2022), Shweta and Yim et al. (2009). This phenomenon signifies an increasing inclination among consumers towards products that are processed naturally, especially in light of the rising prevalence of diseases. Our results are in akin with the previous findings as in our survey also the majority of participants preferred organic incense sticks over synthetic one. Similar investigation was reported to examine how agricultural by products can be transformed into value-added products, highlighting the environmental benefits (Ghosh and Bhattacharya, 2018). The investigation results from Dr. YS Parmar University of Horticulture and Forestry (2023) experimented with various combinations of flower petal powders and adhesives. The best mix for optimal results was **rose + marigold petals (each ~42.5 %) with jigat powder (15 %)**, offering good burning characteristics. Our results are well in akin to the above reports. By concentrating on the formulation of organic incense sticks from flower litters derived from rose (Rosa L.) and *Chrysanthemum*, (*Chrysanthemum* x *Chrysanthemum*) Our research corresponds with this shift in consumer preferences as the flower litters were reported to show no harmful emissions such as CO_2, NO_2 or volatile organic compounds,thereby reassuring **healthier and eco-friendlier alternative** to conventional incense sticks Kalyankar (2024). Furthermore, qualitative phytochemical analyses were performed on the above mentioned flower litters in order to investigate their capacity as biodegradable substances (refer to Table 3). Flower litter were reported to show faster decomposition and relatively higher N,P,K and water holding capacity as compared to other plant part litters. Alencar et al., (2024) Our investigations depicted similar response through their qualitative analysis,these results indirectly prove their benefits in the means of better combustable substance. An additional insights into the environmental impact of converting floral waste into useful products was also reported (Kumar and Singh, 2019). The rose flower powder has a higher amount of carbon (66.98 %) and yellow *Chrysanthemum* flower contains less carbon (37.68 %) compared to other crushed flowers (Kiran and Kavitha, 2023). Our research reassured the eco-friendly properties of flower litter, Additionally the flower litters have shown to have unique biogeochemical dynamics through nutrient and carbon recycling properties (Alencar et al., 2024).

The qualitative analysis reveals significant differences in the phytochemical composition of flower litter 1 (Rosa L.) and flower litter 2 (*Chrysanthemum* x *Chrysanthemum*), indicating that these two types of litter have varying nutrient contents and potential for decomposition. Significantly, the moisture content and conductivity of flower litter 2 are marginally greater than those of flower litter 1, suggesting possible discrepancies in microbial activity and modulation of soil fertility. Divergences in key parameters including nitrogen content, total organic carbon, ammonia, and the C:N ratio additionally indicate variations in the availability of organic matter and the dynamics of nutrients. These factors have the potential to impact decomposition processes and nutrient cycling within ecosystems. Furthermore, the heterogeneity of micronutrient concentrations, such as magnesium, iron, and manganese, underscores the potential advantageous properties of these floral litters in terms of augmenting soil fertility and sustaining plant nutrition (Alencar et al., 2024). Furthermore, our examination of the formulated incense sticks provides significant observations regarding their tangible attributes and commercial viability, as succinctly outlined in Table 6.

To examine the texture, burning longevity,sniff quality and binding capacity with stick, different combination of burning and binding material was tried out. Charcoal, Charcoal and Flower litter 1, Charcoal and Flower litter 2, Sawdust, Sawdust and Flower litter 1,Sawdust and Fower litter 2 and Cow dung was chosen as the Burning material, Two different Binding material was used, wheat paste and Jigat. Among the combination Flower litter 1 with Jigat had excellent Burning longevity,Sniff and Texture.Wheat paste performed well with Flower litter 1 and charcoal combination. The detailed results were depicted in Table 6. Diverse textures, fragrances, and burning duration were observed in the end products produced by combining flower litters with various binding and combustion substances. The observation suggest that Flower litter 1 (Rosa L.) performed well on wheat paste formulation in context with texture, fragrance, longevity of combusition etc. It is worth noting that charcoal-based formulations demonstrate enhanced smoldering longevity and aroma in comparison to sawdust-based formulations. The probable reason may be because of the higher carbon content, lower moisture and more compact structure of the charcoal. This underscores the importance of material selection in the optimization of product performance. In addition, the significance of aromatic properties and health benefits in shaping consumer preferences is illuminated through regression analysis, as shown in Table 2b.

The market potential of organic incense sticks is substantiated by our statistical analyses, which are presented in Tables 1, 2a, and 2c. The ANOVA outcomes highlight notable discrepancies among synthetic and organic incense sticks with respect to multiple parameters, thereby underscoring the unique qualities of natural alternatives. Post hoc testing utilizing Tukey's HSD provides additional validation for distinct pairwise distinctions among the brands, thereby strengthening consumer perceptions regarding quality and preference. The results of this study validate previous research, thereby affirming the marketability and favorability of organic incense sticks among consumers. Many startups were reported in Delhi,Kanpur and Coimbatore which converts floral waste into incense sticks (The Hindu Bureau, 2024; Sharma, 2024; Nirmalaya, 2024). The expanding scope of floral biomass beyond incense sticks includes vegan leather and biodegradable packaging materials (Raj, 2025). We provide a comprehensive comprehension of the value proposition and potential impact of natural alternatives in the incense stick industry through the integration of meticulous scientific analysis and market insights.

5. Conclusion

The market potential for formulated organic incense sticks exhibits a positive trajectory, as evidenced by a modest improvement when compared to synthetic brands. The results mentioned above serve as encouraging indicators for the potential market viability of further organic products. Through the integration of processed floral byproducts into the composition of incense sticks, we have conducted comprehensive experiments involving various combinations of burning and binding substances to assess their textural qualities, combustion duration, and aesthetic appeal. Incense sticks crafted from dehydrated rose petals exhibit improved characteristics with regard to texture, fragrance, and duration of combustion, specifically when wheat paste is

Table 6
Qualitative analysis of Incense sticks formulated with flower litter 1 and 2 along with different varieties of burning and binding material.

Ingredients			Observation				Statistical analysis		
Burning Material	**Binding material**	**Fragrance**	**Texture**	**Burning longevity**	**Sniff quality**	**Binding with stick**	**ANOVA p-value**	**Regression p-value**	**Tukey's Test p-value**
Charcoal	wheat paste	Rose or	Smooth	21.13	Better	Yes	0.015	0.002	0.002
charcoal+ flower 1	wheat paste	Jasmine Oil	Smooth	34.19	Good	Yes	0.023	0.001	0.001
charcoal+ flower 2	wheat paste		Rough	31.07	Good	Yes	0.018	0.003	0.003
saw dust	wheat paste		smooth	14.22	Better	No	0.041	0.005	0.005
Saw dust + f1	wheat paste		smooth	19.42	Good	No	0.037	0.004	0.004
Saw dust + f2	wheat paste		rough	17.15	Good	No	0.046	0.006	0.006
flower 1	wheat paste		smooth	0.12	Best	NO	0.055	0.009	0.009
flower 2	wheat paste		rough	0.9	Best	NO	0.049	0.007	0.007
Charcoal	Jigat		smooth	25.03	Better	YES	0.028	0.001	0.001
charcoal+ F1	Jigat		smooth	48.27	Good	YES	0.011	0.000	0.000
charcoal+ F2	Jigat		smooth	42.57	Good	YES	0.009	0.000	0.000
saw dust	Jigat		smooth	35.47	Better	YES	0.030	0.001	0.001
Saw dust + f1	Jigat		smooth	40.45	Good	Yes	0.010	0.000	0.000
Saw dust + F2	Jigat		smooth	39.37	Good	Yes	0.008	0.000	0.000
flower 1	Jigat		smooth	50.12	Excellent	YES	0.005	0.000	0.000
flower 2	Jigat		smooth	0.20	Excellent	YES	0.034	0.002	0.002
Charcoal+f1 +f2 + saw dust	Jigat		Smooth	38.03	Excellent	YES	0.012	0.000	0.000
Cow dung	Jigat		Rough	0.10	Better	Yes	0.052	0.008	0.008
Cow dung + F1	Jigat		Rough	0.11	Good	Yes	0.048	0.007	0.007

integrated,as the denser nature of wheat paste provides better binding of the volatile oils and thereby prolongs the combustion duration. Further investigation on the binding material is essential as wheat being one of the staple food, a better alternative (Agricultural or Industrial waste product) binding product can be found. It is worth mentioning that when utilized in isolation, the bonded material jigat transforms from a silky consistency to a granular one when combined with a combustible substance, such as bovine manure. Moreover, the positive results obtained from the survey conducted to assess the market potential of organic incense sticks serve as evidence of the growing preference for natural products.

The recognition of a multitude of biodegradable phytoconstituents in floral byproducts via phytochemical analysis enhances the appeal of organic incense sticks as environmentally sustainable alternatives. In contrast to synthetic incense sticks, organic incense sticks decompose entirely and do not contribute to air pollution. Hence, incense sticks crafted from floral byproducts, particularly rose (Rosa L.) and *Chrysanthemum*, (*Chrysanthemum* x *Chrysanthemum*) exhibit superior physical and chemical properties in comparison to synthetic alternatives. Moreover, we add to the management of solid waste and produce products with increased value by incorporating flower debris into our processes. Opportunities abound for nascent industries in both rural and urban areas as a result of the favorable market potential of organic incense sticks. In brief, our study underscores the functionality and desirability of organic incense sticks, thereby proposing a feasible and environmentally conscious substitute that exhibits substantial potential for commercialization. As consumer preferences continue to shift in favor of natural and environmentally sustainable alternatives, the organic incense stick industry is poised for significant growth and development.

CRediT authorship contribution statement

Sekar Malathy: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Dhanraj Dr. k. Manoj:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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